

Charan Vardham

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EDUCATION

University of Wisconsin Madison

Madison, WI

Bachelor of Science in Computer Science, Minor in Mathematics - 3.9 GPA (CWRU transfer) Aug. 2024 – April 2028

- **Relevant Courses:** Data Structures, Algorithms, Discrete Mathematics, Linear Algebra, Computer Security

EXPERIENCE

Open Source Contributor

May 2026 – Present

PyTorch

Remote

- Fixed **3** bugs in PyTorch Inductor's `torch.compile()`, achieving a **2x** improvement in graph compilation speed
- Added CPython's `descr_check` validation to Dynamo's C-descriptor variable tracking, fixing incorrect control flow during tracing when incompatible types bind to C descriptors

Undergraduate Student Researcher

Sep 2026 – Present

University of Wisconsin Madison

Madison, WI

- Implementing a differential static timing analysis engine in **C++** on top of OpenTimer, computing gradients of timing quantities with respect to design parameters
- Building **reverse mode automatic differentiation** with operation tracking to record and replay the timing propagation graph for gradient computation
- Using Taskflow task graph parallelism to run timing propagation and backward passes concurrently across large circuit graphs

xLab Intern

Jan 2026 – May 2026

Weatherhead School of Management, CWRU

Cleveland, OH

- Built an AI case study platform using LLMs for business simulation, serving **200+** students across **3** MBA courses
- Deployed a full-stack application with Next.js (TypeScript), Tailwind CSS, PostgreSQL, and Prisma

AI/ML Intern

May 2025 – Aug 2025

IQVIA

Remote

- Built a scalable RAG-based medical chatbot processing **200MB+** of clinical documents to reduce hallucinations and improve factual accuracy
- Implemented document ingestion, chunking, embedding, and retrieval pipelines, achieving **92% Recall@5** and **0.81 MRR** on a manually labeled set of 200 clinical queries
- Worked on a separate voice-enabled chatbot project, integrating NVIDIA Riva APIs for real-time STT/TTS with an LLM backend

PROJECTS

GPT-2 Pretraining ([link](#)) | *PyTorch, Python, CUDA, SLURM*

February 2026

- Trained GPT-2 (**124M** parameters) from scratch on **10B** tokens using NVIDIA H100 GPUs, achieving final loss of **3.0** with a throughput of **260K** tokens/sec
- Implemented a custom **FlashAttention-2** kernel in **Triton** that resulted in a **10-50%** throughput improvement.

Real-Time Credit Card Fraud Detection ([link](#)) | *Python, TypeScript, XGBoost, AWS, Docker*

June 2026

- Built an end-to-end ML fraud detection system using a stacking ensemble model trained on the **IEEE-CIS dataset** and deployed in a Docker container on ECS Fargate via SST
- Implemented a **Next.js 15 inference API** (predict, transaction history, health check) with TypeScript types shared across the stack, writing predictions to DynamoDB on every request
- Designed CI/CD on **GitHub Actions** with TypeScript type-checking, Python linting, and automated SST deployment to AWS

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, R, HTML/CSS, C++

Frameworks / Libraries: React, Node.js, Next.js, FastAPI, Material-UI, Pandas, NumPy, Matplotlib, Scikit-Learn, PyTorch, TensorFlow, PyTorch Lightning, Hugging Face Transformers

Developer Tools / Deployment: Git, Docker, VS Code, IntelliJ, PyCharm, Visual Studio, REST APIs, AWS, CI/CD, Jenkins, Kubernetes, Unix, Azure, Google Cloud, Grafana, Firmware

Testing / ML Skills: Unit testing, Data preprocessing, Feature engineering, Model evaluation metrics (PSNR, SSIM), RAG pipelines, LLM integration, Load Balancing, Consensus Algorithms, Bash, Prometheus